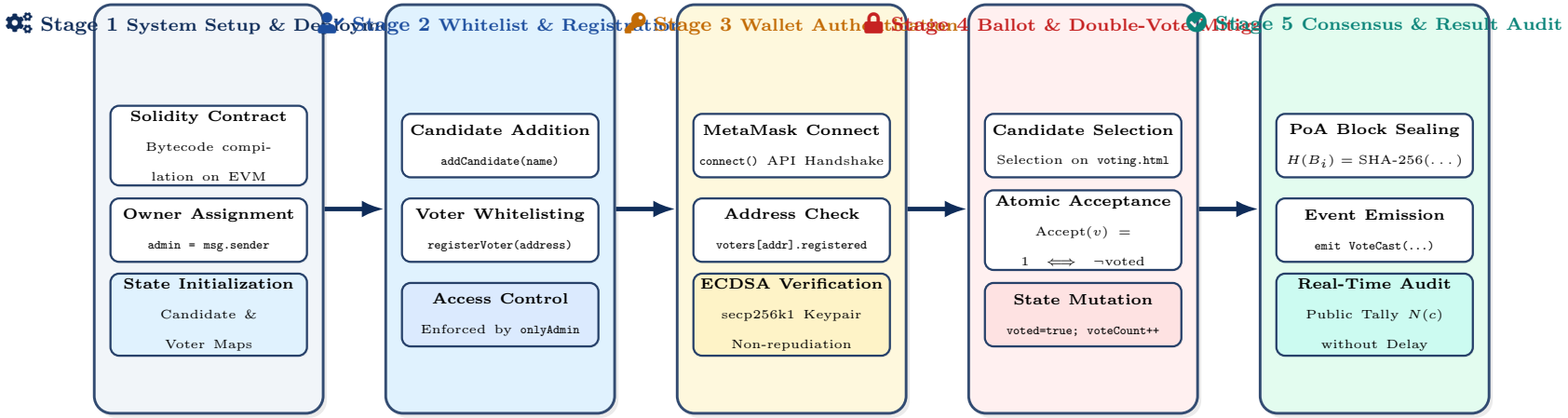

Stage 1 System Setup & Deployment
Solidity Contract
Bytecode compi-
lation on EVM
Owner Assignment
admin = msg.sender
State Initialization
Candidate &
Voter Maps
Stage 2 Whitelist & Registration
Candidate Addition
addCandidate(name)
Voter Whitelisting
registerVoter(address)
Access Control
Enforced by onlyAdmin
Stage 3 Wallet Authentication
MetaMask Connect
connect() API Handshake
Address Check
voters[addr].registered
ECDSA Verification
secp256k1 Keypair
Non-repudiation
Stage 4 Ballot & Double-Vote Prevention
Candidate Selection
Selection on voting.html
Atomic Acceptance
Accept(v) =
1 ⟺ ¬voted
State Mutation
voted=true; voteCount++
Stage 5 Consensus & Result Audit
PoA Block Sealing
H(B_i) = SHA-256(...)
Event Emission
emit VoteCast(...)
Real-Time Audit
Public Tally N(c)
without Delay